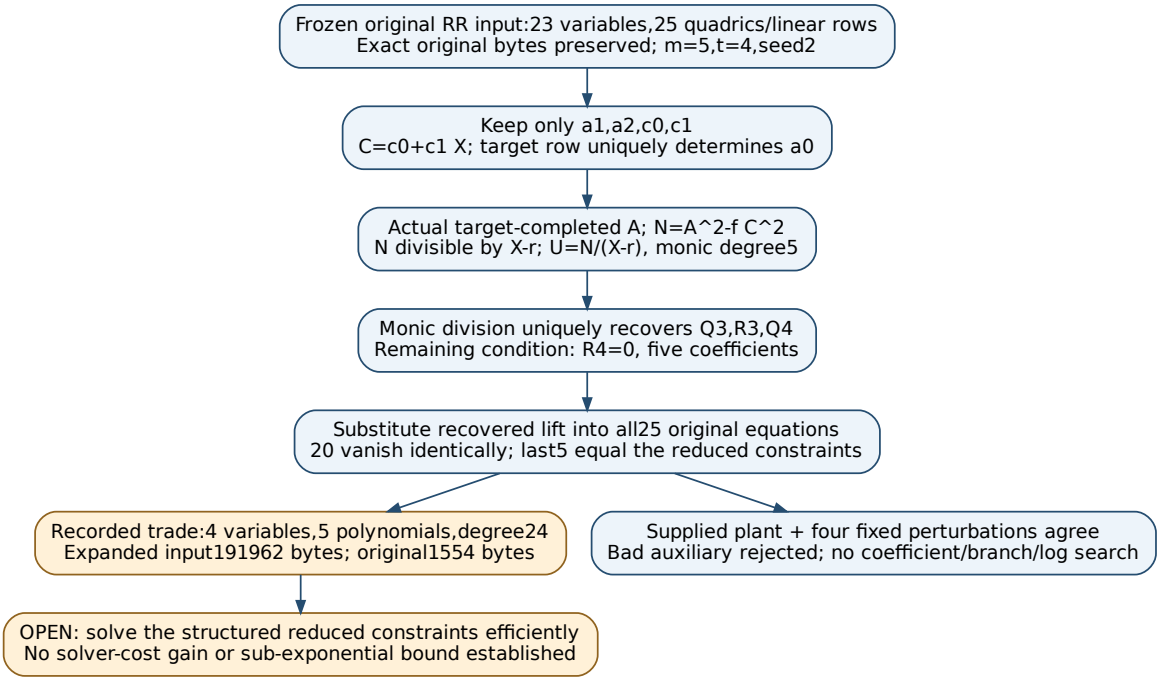
Lossless projection of the unfinished m5 norm input

2026-11-13. Source`ccd9e23f1`. Unsigned evidence [pcc:unverified].

The actual23-variable,25-equation RR input reduces to four function coefficients. The other19 variables are uniquely recovered by target completion and monic polynomial division. Native symbolic substitution checks all original equations:20 vanish identically, the last5 are exactly the reduced membership constraints. This is a lossless representation trade, not a measured solver gain [pcc:unverified].



For $C = c_0 + c_1 X$, $D = X^3 + a_2 X^2 + a_1 X$, set

$$a_0 = zC(r) - D(r), \quad A = D + a_0, \quad N = A^2 - fC^2, \quad U = N \operatorname{div} (X - r).$$

The actual target satisfies$z^2 = f(r)$; the norm division is exact and U is monic degree5. No variable denominator or new saturation/chart guard discards old algebraic solutions. The universal Lean model and concrete native compiler binding are separate evidence scopes [pcc:unverified].

Unsigned claims: canonical quotient`PCC-20261113-c1ec0b`; terminal membership`PCC-20261113-a3e4cb`; native binding`PCC-20261113-b465d8`.

Exact controls and the degree/input trade

The copied Dickson m5,t4,seed2 fixture has$p = 14680063$, $f = X^3 + 3607823X + 6826282$, target$(r, z) = (5400989, 8855456)$, terminal$c = 12691281$. Original input SHA256:

`9ba376d81c05109b11102c645153f9204df233600fec5851993a6bd5787e3cff`. Exact original bytes and historical paths remain intact [pcc:unverified].

Let$D_0 = X$, $D_{j+1} = D_j^2 - 2$. Compute the actual divisions

$$D_3 = UQ_3 + R_3, \qquad R_3^2 - 2 - c = UQ_4 + R_4.$$

Only the five coefficients ofR_4 remain. The direct remainder $(D_4 - c) \bmod U$ equalsR_4. Every one of the25 original source equations is parsed independently and substituted into exact four-variable sparse polynomials. The first20 become zero polynomials; the last5 equal those five residuals coefficient by coefficient [pcc:unverified].

Reduced$X$ coefficient	Total degree	Terms
4	24	1103
3	24	1445
2	24	1830
1	24	2171
0	24	2366

The expanded reduced input is191962 bytes; the original is1554 bytes. Both full inputs are retained. The generator records682901 sparse coefficient-pair products and peak2366 observed terms below the finite guards. These are construction counts, not a solver cost model or timing. The one write-once artifact exits0 under256MiB/120 CPU-second caps [pcc:unverified].

The five prospectively fixed assignments are the supplied plant plus one increment of each core coordinate. The plant recovers all23 supplied values and all25 residuals vanish. Each perturbation preserves the20 identically zero rows, matches the reduced five residuals exactly and is nonzero in this fixture. Corrupting the recovered Q3 leading coefficient rejects. No search for coefficients, branches, roots or logarithms; no Groebner solver or paid job ran. Unit tests reuse these cases [pcc:unverified].

Raw-source audit. The earlier m5,t4 RR comparison never completed. R2b timed out; R3 RR attempts were interrupted. RRa2's exact input hash matches, its last log begins an unfinished464562-by390983 degree7 matrix, and the raw CloudTrail object records BidEvictedEvent at11:09:30Z on Nov9. The protocol grants no bounds from Spot-lost partial jobs. The object has no pod/node field; that association relies on the historical assessment. Idle-start msolve output and earlier close CPU/real pairs do not establish continuous isolation or final-step timing. A reliable retry could decide one cell, not an asymptotic growth law [pcc:unverified].

Source audit`PCC-20261113-08f893`; copied source paths, hashes, raw metadata and explicit limits are in`sources-checked.json`.

Universal model, actual source binding and next test

Lean4.35.0.

CryptoLean `148f5dc389b341401e06d76abe7783c4ccf42f2e`.

Mathlib `cfc55fc58714a59dc06e6e79edc6d1d95140d3c2`.

Build004 passes16 new statements with only propext, Classical.choice and Quot.sound. Current source gate exits0:286 statements,0 additional trusted. The new module's import-inclusive fresh replay76692 exits0. Native tests002 pass3, strict clippy002 and release build pass. Failed attempts remain. All scientific claims are unsigned; default-PATH PCC checks remain source-only not_covered [pcc:unverified].

NormMembershipElimination	Exact statement scope
completionRow / completionUnique	Actual constant completion; target row and uniqueness
normAtTarget / canonicalNormQuotient	Actual target/curve equations; norm zero and exact computed quotient
normQuotientUnique / divisionUnique	Actual monic factor, quotient and bounded remainder are determined
reducedStep / reducedMaster	Actual modular tower equals direct master remainder
canonicalTowerLift / canonicalTowerBound	Actual computed auxiliaries exist with the degree guard
towerUnique	Every bounded old tower equals the computed one
terminalMembership	Nontrivial ring, positive-degree monic modulus; terminal iff divisibility
normWindow / supportWindow	Field, monic cubic$A$, cubic$f$, $\deg B \leq 1$; norm/support monic degrees6/5
completionWindow / quotientWindow	Completion preserves degree3; quotient degree≤ 3

The model does not assume successful old auxiliaries to construct the new ones. Its universal monic-division facts are distinct from the native normal-form binding of all25 literal expressions. No native comparison is presented as a Lean-verified compiler [pcc:unverified].

Literal certificate`IDC1hb49f2953ff1e646f`, attestation `PCC-20261113-5e61c7`, covers only$(D + ZB - D)^2 - Z^2B^2 = 0$. The unchanged canonical checker accepts32+16 replay points over its v1 fixed modulus. It does not encode monic division or the full projection; the Lean pointer is separate. Its driver builds, but strict clippy retains four existing style errors in unchanged canonical SHA256 under Rust1.99.1, without suppression or checker edits [pcc:unverified].

Sources checked. KB MCP/executable unavailable. Read named public repo sources and raw records, PKC2016 sections3.1–3.5/4.1/Table4, Cheng cs0507026 Lemma3 and Caminata–Gorla2112.05579v2 section4. Binary Nagao coefficient/block work is adjacent. Query token matches and varying revisions do not establish exhaustive absence or novelty. Two ePrint full texts returned403 and remain algorithmically unassessed [pcc:unverified].

Next discriminating action. Compare a module-aware or circuit-preserving solver on this exact projection with the strongest matched original system, counting construction and every solving phase. Four variables and degree24 alone establish neither a gain nor a loss. No such solver/cost experiment ran. The actual finder and completeexp$(o(\log r))$ bit-cost bound remain open; no benchmark/leaderboard/performance graph values change [pcc:unverified].